# Solar ESS Company Limited

**Project scope – combined PV carport + low PV structure** · 42 × LONGi Hi-MO X10 650 W = 27.30 kWp · gutter (Rev3) · carport floor (Rev4) · aluminium mid clamps (Rev7) · 10 mm gaps both ways, TX-10 (Rev9)

**B0**

Dimensions in metres (details in mm) · print A3 for scale
Preliminary – confirm with a licensed civil / structural engineer

| Installed capacity | PV structures | Cement floor | Pebble ground |
|---|---|---|---|
| **27.30 kWp** | **28 + 14** | **99 m²** | **162 m²** |
| 42 × LONGi Hi-MO X10 Scientist 650 W bifacial | carport (6 cars) + low PV structure – panels | under the PV carport only (S2) | 1.50 m band all round + low-structure zone |

## Scope of works

| # | Item | Scope | Panels | kWp |
|---|---|---|---|---|
| 1 | **PV modules** | 42 × LONGi Hi-MO X10 Scientist 650 W bifacial (LR7-72HVD-650M) = 27.30 kWp · 3 strings of 14, all facing south | 42 | 27.30 |
| 2 | **PV carport structure** | 28 panels · 2.30 / 2.50 m · 3 frames at 5.70 m · 6 cars · purlins 4"×2"×4.0 + aluminium mid / end clamps on M8 rivet nuts (page 4, D1) | 28 | 18.20 |
| 3 | **Low PV structure** | 14 panels (7 + 7) · 1.50 / 1.70 m · 2 frames 4.0 m apart · purlins 4"×2"×2.3 + aluminium mid / end clamps on M8 rivet nuts (page 4, D1) | 14 | 9.10 |
| 4 | **Carport floor + pebble ground** | Cement floor under the carport only 99.1 m² (120 mm, 280 ksc, mesh) + pebbles 161.7 m² incl. under the low structure + edging (sheets F1–F3) | – | – |
| 5 | **Rain gutter + downpipe** | Carport front edge: box gutter 200 × 125 mm, 16.14 m, falls to the west · Ø100 PVC downpipe under the west edge to a post at the back (sheet G1) | – | – |
| 6 | **PV roof waterproofing** NEW | Thin EPDM T-seal TS-10 in every 10 mm module gap (long sides and short ends), 13 moulded crosses TX-10 and 52 clamp boots TB-10 (sheets W1–W4) | – | – |
| | **TOTAL** | | **42** | **27.30** |

### Included

- 42 PV modules LONGi Hi-MO X10 Scientist 650 W bifacial (delivered).
- Steel carport (28 panels, 6 cars) and low PV structure (14 panels): steel, 4"×2" purlins, 60 + 32 aluminium mid / end clamps on M8 rivet nuts, footings, earthing, paint, labour.
- Site preparation, cement floor under the carport 99.1 m², pebble ground 161.7 m² (incl. under the low structure), edging, joints.
- Carport rain gutter along the front (south) edge draining to the west, Ø100 downpipe to the back (north-west) on its own post, splash block on the pebbles (sheet G1).
- Waterproof PV carport roof: TS-10 in all 10 mm module gaps, TX-10 crosses and TB-10 clamp boots, laid and water-tested (sheets W1–W4).

### Not included

- Inverter, DC/AC cabling, protection and metering; panel installation labour.
- Inverter room (built by others), PEA grid connection and permits.
- Sealing the low PV structure (option, W2) – it stands on pebbles with no cars under it.

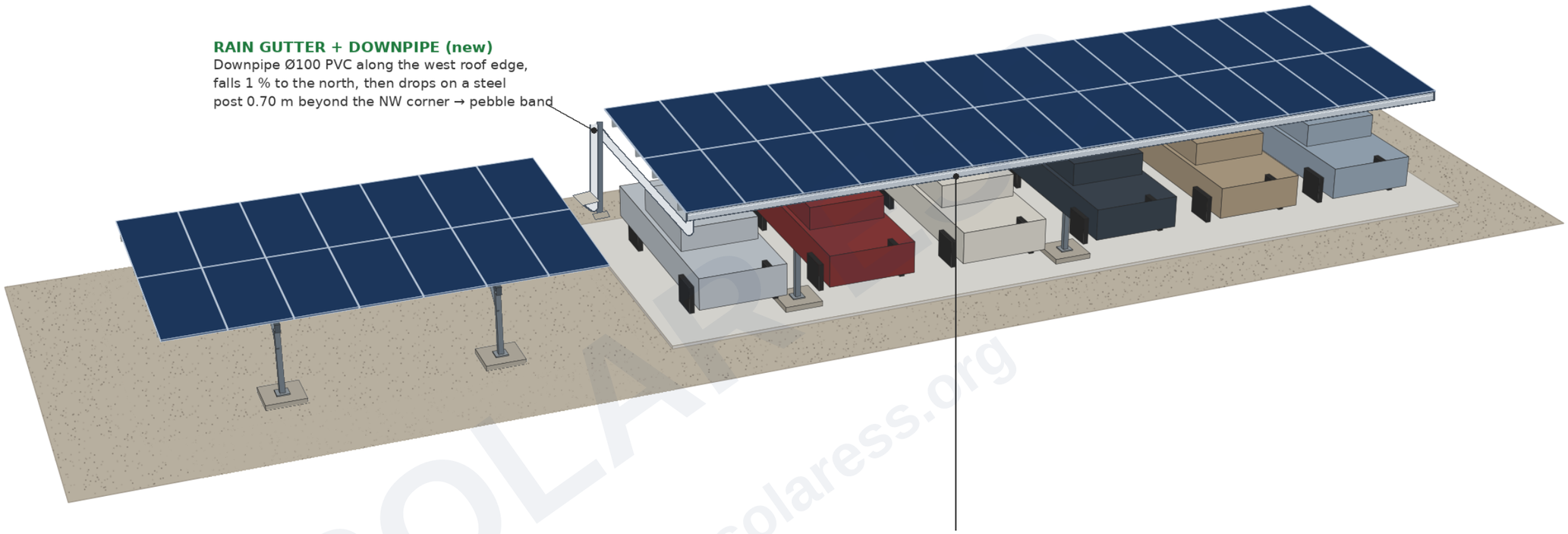
SOLAR ESS www.solaress.org

# Solar ESS Company Limited

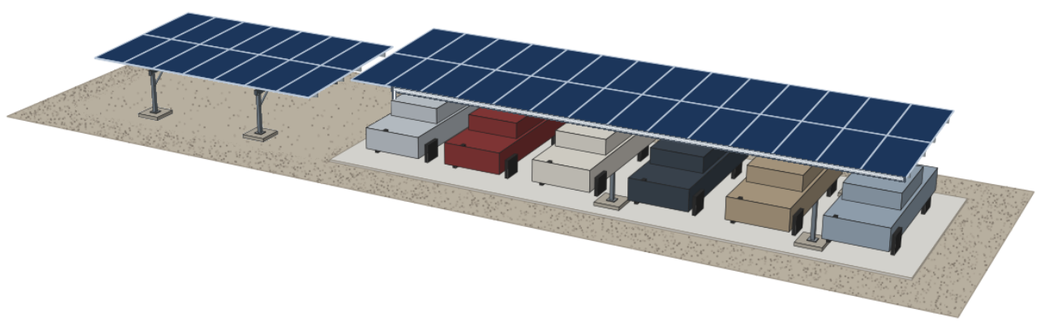
Illustrative 3D model

Combined design – 42 × LONGi Hi-MO X10 650 W (27.30 kWp), facing south · carport 28 panels (2.30/2.50 m, 6 cars) + low PV structure 14 panels (1.50/1.70 m) · cement floor under carport only, pebbles elsewhere

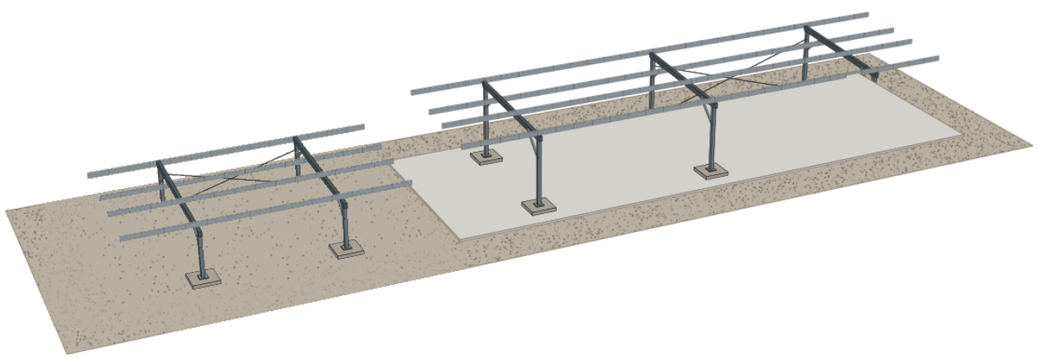
## 3D VIEW – from south-west: low PV structure (14) on pebbles · PV carport (28) on a cement floor · pebble ground all round




## From south-east



## Steel only – panels and cars hidden



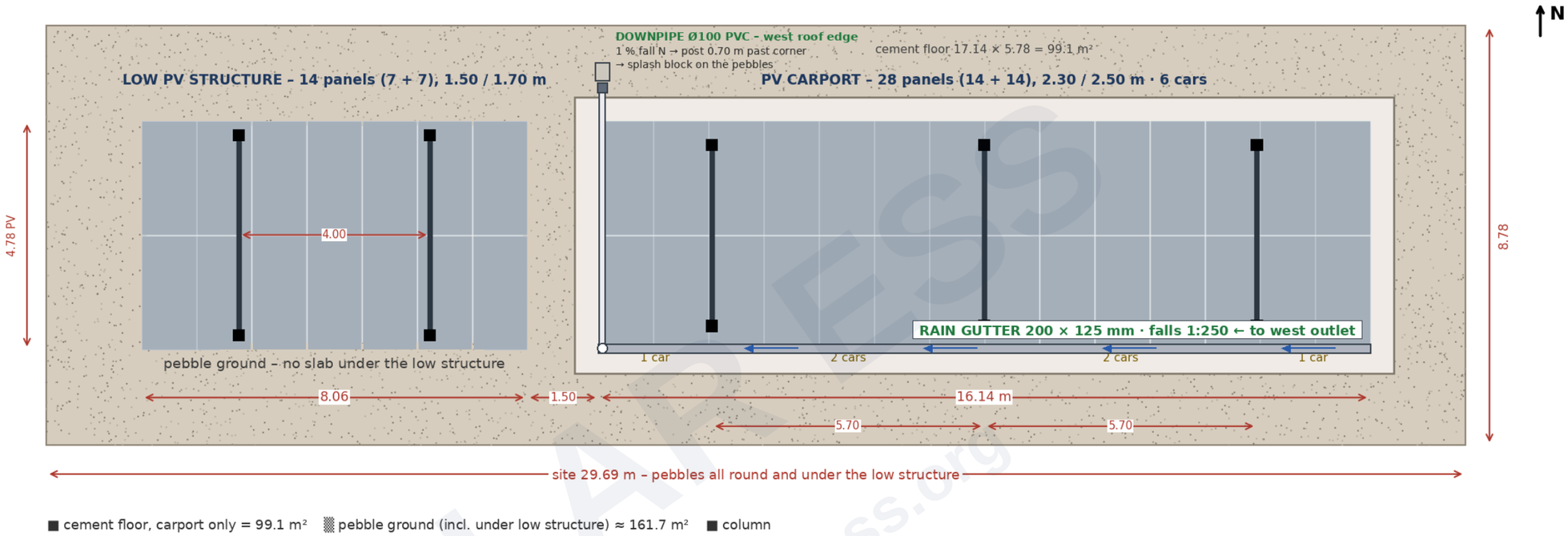
SOLAR ESS
www.solaress.org

# Solar ESS Company Limited

Combined design – carport 28 panels + low PV structure 14 panels (42 × 650 W = 27.30 kWp) – plan, elevation and floors

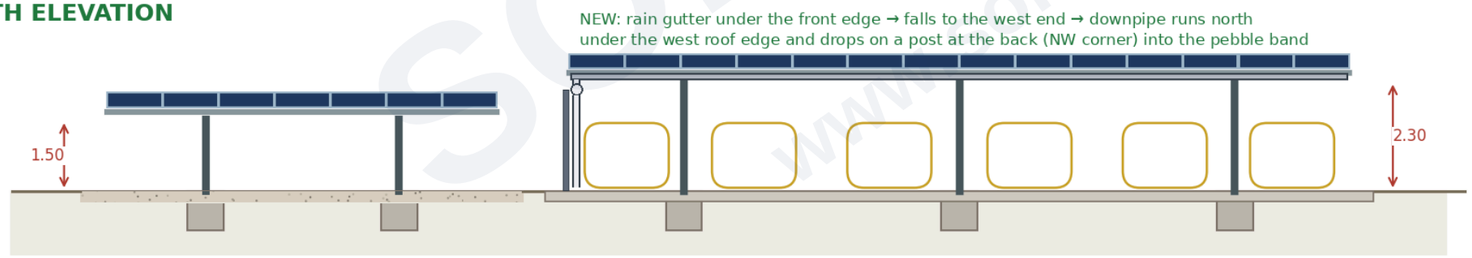
Dimensions in metres
Illustrative – confirm by licensed engineer

## SITE PLAN – all panels facing south · no inverter room · cement floor under the carport only (sedans & pickups) · pebbles elsewhere

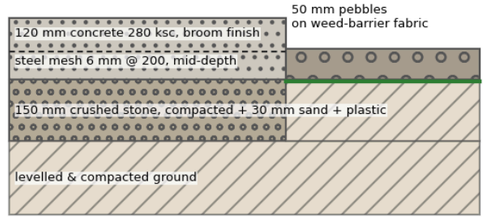



■ cement floor, carport only = 99.1 m² ▒ pebble ground (incl. under low structure) ≈ 161.7 m² ■ column

## SOUTH ELEVATION




## FLOOR SECTION – sedans & pickups




- Same structure type for both PV roofs: 4"×4" columns, 6"×4" beams with knee braces, 4"×2" purlins on edge (carport 4.0 mm, low 2.3 mm) + aluminium mid / end clamps on M8 rivet nuts (Rev7, D1), RB16 rod bracing, footings 0.75 × 0.75 × 0.8 m.
- Carport: 2 × 14 panels, 10 mm gaps both ways – roof 16.01 × 4.77 m (Rev9; plan above drawn with the old 20 mm gaps), 3 frames at 5.70 m, purlins cantilever ≈ 2.30 m each end – 2 cars per bay + 1 under each cantilever = 6 cars.
- Low PV structure: 2 × 7 panels (no spare slot), roof 8.00 × 4.77 m (Rev6), 2 frames 4.00 m apart, columns 4.18 m front–back, purlins cantilever ≈ 2.00 m; 1.35 m clear under the front beam.
- Strings: 3 × 14 (carport 2, low structure 1) – all panels face south at ≈3°. No inverter room in this design – the inverter location is agreed separately.
- Floor for sedans & pickups – carport only: 120 mm slab 280 ksc, mesh 6 mm @ 200, on 150 mm crushed stone, 0.50 m beyond the carport roof (99.1 m²), thickened edges; saw-cut joints on column lines (≤ 2.9 m bays); pebbles 161.7 m² incl. under the low structure – F1–F3.

SOLAR ESS
โซลาร์ เอสส์

Solar Energy for Sustainable Solution

# Combined design – structures summary (42 × LONGi Hi-MO X10 650 W = 27.30 kWp)

| Item | Panels | kWp |
|---|---|---|
| PV carport, 28 panels, 2.30 / 2.50 m, 6 cars · 4"×2" purlins + aluminium mid / end clamps | 28 | 18.2 |
| Low PV structure, 14 panels, 1.50 / 1.70 m (same design, 4"×2"×2.3 purlins) | 14 | 9.1 |
| Floor works – cement floor under the carport only 99.1 m² (120 mm, 280 ksc) + pebble ground 161.7 m² | – | – |
| **TOTAL** | **42** | **27.3** |

**Rev7 – aluminium mid / end clamps on M8 rivet nuts (D1) · thin EPDM seals, boots and crosses (W1–W4)**

Carport front gutter, falls to the west outlet, Ø100 downpipe to the NW post (sheet G1)

Cement floor under the carport only; the low PV structure stands on pebbles (sheets F1–F3)

Inverter room removed from this design (built by others). Excludes PV modules, inverter, cabling and permits.

## PV carport – 28 panels, 2.30 / 2.50 m, 3 frames at 5.7 m, 6 cars · material quantities

| # | Item | Specification | Qty | Unit |
|---|---|---|---|---|
| 1 | Columns | Tube 4"×4" × 2.3 mm, 6 m (front + back column per length) | 3 | length |
| 2 | Beams | Tube 6"×4" × 2.3 mm, 6 m (≈3.9 m beam) | 3 | length |
| 3 | Knee braces | Tube 2"×2" × 2.3 mm, ≈0.85 m, 2 per frame | 1 | length |
| 4 | Purlins / PV rails (4 lines) | Tube 4"×2" × 4.0 mm (100×50, on edge), 6 m – bolted splices | 12 | length |
| 5 | Purlin splice plates | 2 side plates 100×8×600 mm per joint – keep clear of clamp positions | 61 | kg |
| 6 | Purlin splice bolts | M12 × 90 through-bolts, 8 per joint | 64 | set |
| 7 | Plates | Base, cap, gusset, end plates, cleats, splice plates | 114 | kg |
| 8 | Anchor bolts | M16 J-bolt L=400 + nuts & washers | 24 | set |
| 9 | Bracing | RB16 rods + turnbuckles, back plane + roof plane | 1 | lot |
| 10 | Purlin bolts | M12 bolts, nuts, washers | 56 | set |
| 11 | Welding consumables | Electrodes, grinding & cutting discs | 1 | lot |
| 12 | Paint | Zinc-rich primer + PU top coat | 1 | lot |
| 13 | Concrete | CPAC 240 ksc, footings + 50 mm lean | 2.9 | m³ |
| 14 | DB12 | DB12 SD40, 10 m (12 verticals per footing) | 8 | bar |
| 15 | DB10 | DB10 SD40, 10 m (screens) | 6 | bar |
| 16 | Binding wire | Tie wire #18 | 1 | kg |
| 17 | Earthing | 2 × earth rod 5/8" × 2.4 m, bonding cable & clamps | 1 | lot |
| 18 | Transport | Delivery of steel & materials to site | 1 | lot |
| 19 | Labour | Excavation, footings, fabrication, erection | 3 | frame |
| 20 | PV mid clamps (Rev7) | Aluminium 6005-T5, 50 mm long, for 30 mm frames, 10 mm gap – 52 + 4 spares (D1) | 56 | pc |
| 21 | PV end clamps (Rev7) | Aluminium 6005-T5, 50 mm long, 30 mm frames – 8 + 2 spares (D1) | 10 | pc |
| 22 | Rivet nuts + bolts (Rev7) | M8 closed-end countersunk steel rivet nut + M8 × 45 A2-70 bolt (D1) | 66 | set |

## Low PV structure – 14 panels (7 + 7), 1.50 / 1.70 m, 2 frames 4.0 m apart · material quantities

| # | Item | Specification | Qty | Unit |
|---|---|---|---|---|
| 1 | Columns | Tube 4"×4" × 2.3 mm, 6 m (front + back column per length) | 2 | length |
| 2 | Beams | Tube 6"×4" × 2.3 mm, 6 m (≈4.3 m beam) | 2 | length |
| 3 | Knee braces | Tube 2"×2" × 2.3 mm, ≈0.85 m, 2 per frame | 1 | length |
| 4 | Purlins / PV rails (4 lines) | Tube 4"×2" × 2.3 mm (100×50, on edge), 6 m – bolted splices | 8 | length |
| 5 | Purlin splice plates | 2 side plates 100×6×600 mm per joint – keep clear of clamp positions | 23 | kg |
| 6 | Purlin splice bolts | M12 × 90 through-bolts, 8 per joint | 32 | set |
| 7 | Plates | Base, cap, gusset, end plates, cleats, splice plates | 79 | kg |
| 8 | Anchor bolts | M16 J-bolt L=400 + nuts & washers | 16 | set |
| 9 | Bracing | RB16 rods + turnbuckles, back plane + roof plane | 0.5 | lot |
| 10 | Purlin bolts | M12 bolts, nuts, washers | 43 | set |
| 11 | Welding consumables | Electrodes, grinding & cutting discs | 1 | lot |
| 12 | Paint | Zinc-rich primer + PU top coat | 1 | lot |
| 13 | Concrete | CPAC 240 ksc, footings + 50 mm lean | 1.9 | m³ |
| 14 | DB12 | DB12 SD40, 10 m (12 verticals per footing) | 5 | bar |
| 15 | DB10 | DB10 SD40, 10 m (screens) | 4 | bar |
| 16 | Binding wire | Tie wire #18 | 1 | kg |
| 17 | Earthing | Bonding to the carport earth system | 1 | lot |
| 18 | Transport | Same delivery trip as the carport | 1 | lot |
| 19 | Labour | Excavation, footings, fabrication, erection | 2 | frame |
| 20 | PV mid clamps (Rev7) | Aluminium 6005-T5, 50 mm long, for 30 mm frames, 10 mm gap – 24 + 2 spares (D1) | 26 | pc |
| 21 | PV end clamps (Rev7) | Aluminium 6005-T5, 50 mm long, 30 mm frames – 8 + 2 spares (D1) | 10 | pc |
| 22 | Rivet nuts + bolts (Rev7) | M8 closed-end countersunk steel rivet nut (grip 1.5–3.0) + M8 × 45 A2-70 bolt (D1) | 36 | set |

## Carport checks

| | |
|---|---|
| Purlin 4"×2"×4.0 span stress, 5.70 m span (S = 28,000 mm³) | 127.1 MPa |
| Purlin span deflection (I = 1.40×10⁶ mm⁴) | 17.9 mm |
| Purlin cantilever stress | 87.7 MPa |
| Purlin cantilever tip deflection | 12.4 mm |
| Worst-frame beam 6"×4"×2.3 stress, with knee braces (S = 50,800 mm³) | 79.4 MPa |
| Footing safety factor | 1.7 – |
| Result | OK |

## Low structure checks

| | |
|---|---|
| Purlin 4"×2"×2.3 span stress, 4.0 m span (S = 17,400 mm³) | 100.7 MPa |
| Purlin span deflection (I = 0.87×10⁶ mm⁴) | 7.0 mm |
| Purlin cantilever stress | 103.6 MPa |
| Purlin cantilever tip deflection | 10.6 mm |
| Worst-frame beam 6"×4"×2.3 stress, with knee braces (S = 50,800 mm³) | 71.5 MPa |
| Footing safety factor | 2.4 – |
| Result | OK |

## Notes

- **PV fixing (Rev7): aluminium mid and end clamps** on M8 closed-end rivet nuts set flush in the purlin top wall; bolts go down the 10 mm module gap, nothing is drilled in the frames. Mid clamps at every joint/purlin crossing (52 + 24), end clamps at the roof ends (8 + 8) – sheet D1. The PSI U-clamps (Rev4) are removed.
- Purlins 4"×2" (100 × 50 mm, stood on edge): carport 4.0 mm wall (5.70 m spans), low structure 2.3 mm. Stresses at or below about 130 MPa, deflection ≤ L/300 – see checks. Panel edge heights stay 2.30/2.50 and 1.50/1.70 m (columns cut 50 mm longer).
- Place purlin splices between clamp positions and keep rivet nuts ≥ 60 mm clear of the splice plates. Don't drill module frames.
- Both PV roofs: 10 mm module gaps both ways (Rev9), width 4.77 m, all 42 panels face south – 3 strings of 14. Carport roof 16.01 × 4.77 m, purlin cantilever ≈ 2.30 m each end, 6 cars; low structure roof 8.00 × 4.77 m, frames 4.0 m apart, shares the carport delivery trip and earth system.
- Preliminary – confirm with a licensed structural engineer.

SOLAR ESS www.solaressco.com

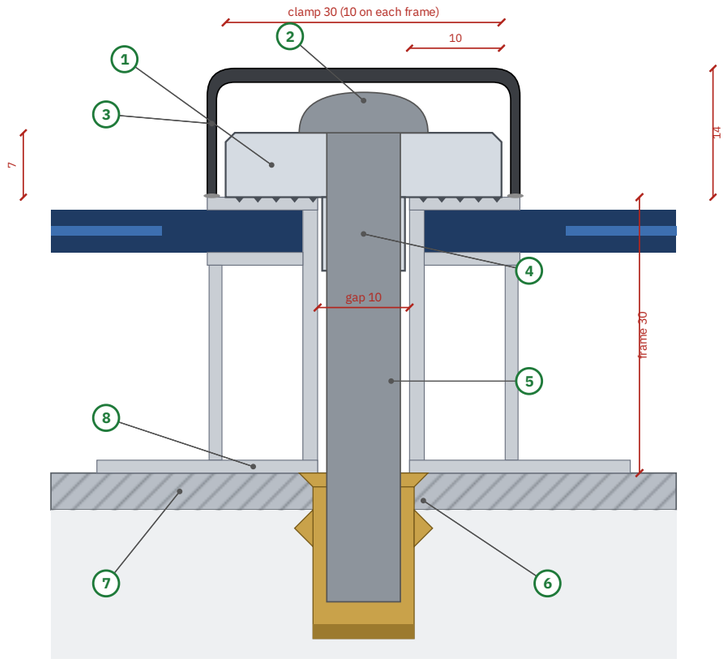
# Solar ESS Company Limited

**PV fixing detail – aluminium mid and end clamps on the 4"×2" purlins with M8 rivet nuts** · 10 mm module gaps

D1

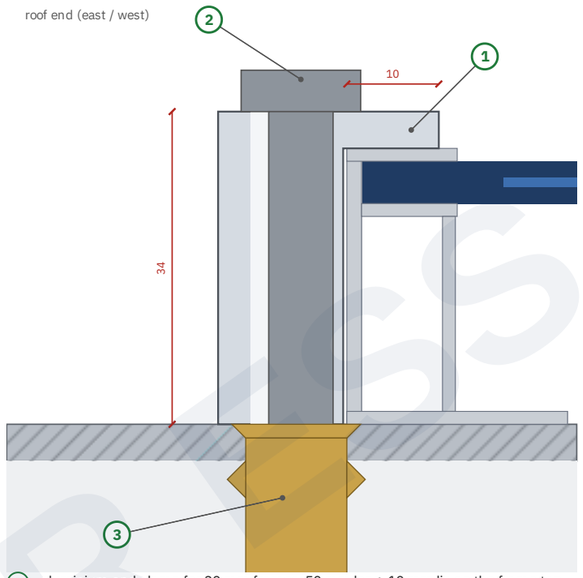
Dimensions in metres (details in mm) · print A3 for scale
Preliminary – confirm with a licensed civil / structural engineer

## SECTION A – across a joint at a mid clamp · scale 1.5:1 on A3 · looking along the purlin




1. aluminium mid clamp, cap 30 × 7 × 50 mm long – 10 mm on each frame (LONGi: 10–12 mm overlap, ≥ 50 mm long, ≥ 4 mm thick)
2. M8 × 45 button-head bolt, stainless A2-70, anti-seize – torque 12–16 N·m (LONGi)
3. TB-10 EPDM boot over the clamp, bedded on MS-polymer (W4)
4. 9 mm web keeps the 10 mm gap and centres the clamp
5. bolt passes down the 10 mm gap – nothing is drilled in the module frames
6. M8 steel rivet nut, countersunk head set flush, closed end – keeps rain out of the purlin
7. purlin 4"×2" top wall – carport 4.0 mm, low structure 2.3 mm; hole Ø 11.0 + 90° countersink
8. frame bottom flanges bear flat on the purlin – nothing stands proud under them

## SECTION B – end clamp at the roof end · scale 1.5:1




1. aluminium end clamp for 30 mm frames, 50 mm long, 10 mm lip on the frame top
2. M8 × 45 socket-cap bolt, A2-70, 12–16 N·m
3. M8 rivet nut, countersunk, closed end – on the purlin top outside the end module

## Quantities

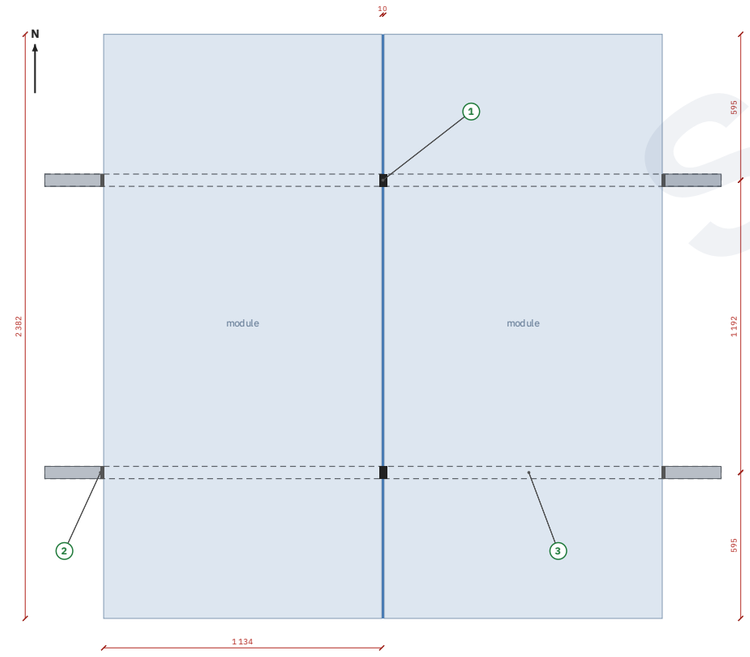
| Item | Carport | Low (option) |
|---|---|---|
| Mid clamps | 52 + 4 | 24 + 2 |
| End clamps | 8 + 2 | 8 + 2 |
| M8 rivet nuts | 60 + 6 | 32 + 4 |
| M8 × 45 A2 bolts | 60 + 6 | 32 + 4 |
| Clamp boots TB-10 (W4) | 52 + 4 | 24 + 2 |

Figures are fitted + spares.

## Installation

1. Mark the clamp points on the purlin tops from the module layout: joint centre lines at 1 144 mm pitch, end clamps outside the end modules.
2. Drill Ø 11.0 mm through the top wall, countersink 90°, deburr, prime the hole edge.
3. Set the rivet nuts with the setting tool; check each one is flush (straightedge) and does not spin.
4. Lay the modules in portrait; set the gaps to 10–11 mm with spacers – never under 10 mm (LONGi).
5. Fit the mid and end clamps with anti-seize on the bolts; torque 12–16 N·m in two passes.
6. Check the earthing bond, then fit the TB-10 boots and the seals (W1–W4).

## PLAN – clamp positions · scale 1:25 · two carport modules




1. mid clamp on every module joint at each purlin – 52 on the carport, 24 on the low structure
2. end clamp at the east and west roof ends – 8 + 8
3. purlins at ≈ L/4 = 595 mm from each module end – keep within the LONGi clamp zone for LR7-72HVD (confirm with LONGi)

## Clamp and fixing specification

| | |
|---|---|
| **Mid clamp** | aluminium 6005-T5, anodised, 50 mm long, cap 30 × 7 mm, 9 mm gap web, serrated underside with earthing pins |
| **End clamp** | aluminium 6005-T5, for 30 mm frames, 50 mm long, 10 mm lip |
| **Overlap** | 10 mm on each frame top – LONGi: 10–12 mm overlap, clamp ≥ 50 mm long, ≥ 4 mm thick, no shading of the cells |
| **Bolts** | M8 × 45 stainless A2-70 – button head (mid), socket cap (end), with anti-seize |
| **Torque** | 12–16 N·m (LONGi, M8) – two passes; re-check after 1 month |
| **Rivet nuts** | M8 steel, zinc-flake, knurled, countersunk head, closed end · grip 3.0–5.0 mm (carport, 4.0 mm wall) / 1.5–3.0 mm (low structure, 2.3 mm wall) · pull-out ≥ 3 kN – confirm with the supplier |
| **Holes** | Ø 11.0 mm + 90° countersink on the joint centre line; deburr, touch up with zinc-rich primer |
| **Positions** | purlins at ≈ 595 mm (L/4) from each module end – within the LONGi clamp zone for LR7-72HVD (confirm) |
| **Earthing** | serrated clamps bite through the anodising; bond each purlin line to the carport earth (page 4) |

## Notes

- Bolts go down the 10 mm gap into the purlin – nothing is drilled in the module frames.
- Closed-end rivet nuts keep rain out of the purlin tubes; set flush so the frames bear flat.
- Keep rivet nuts ≥ 60 mm clear of purlin splice plates; place splices between clamp positions.
- Rev7: the PSI U-clamps (Rev4) are removed from the design. The 4"×2" purlins and the checks on page 4 stay as they are.

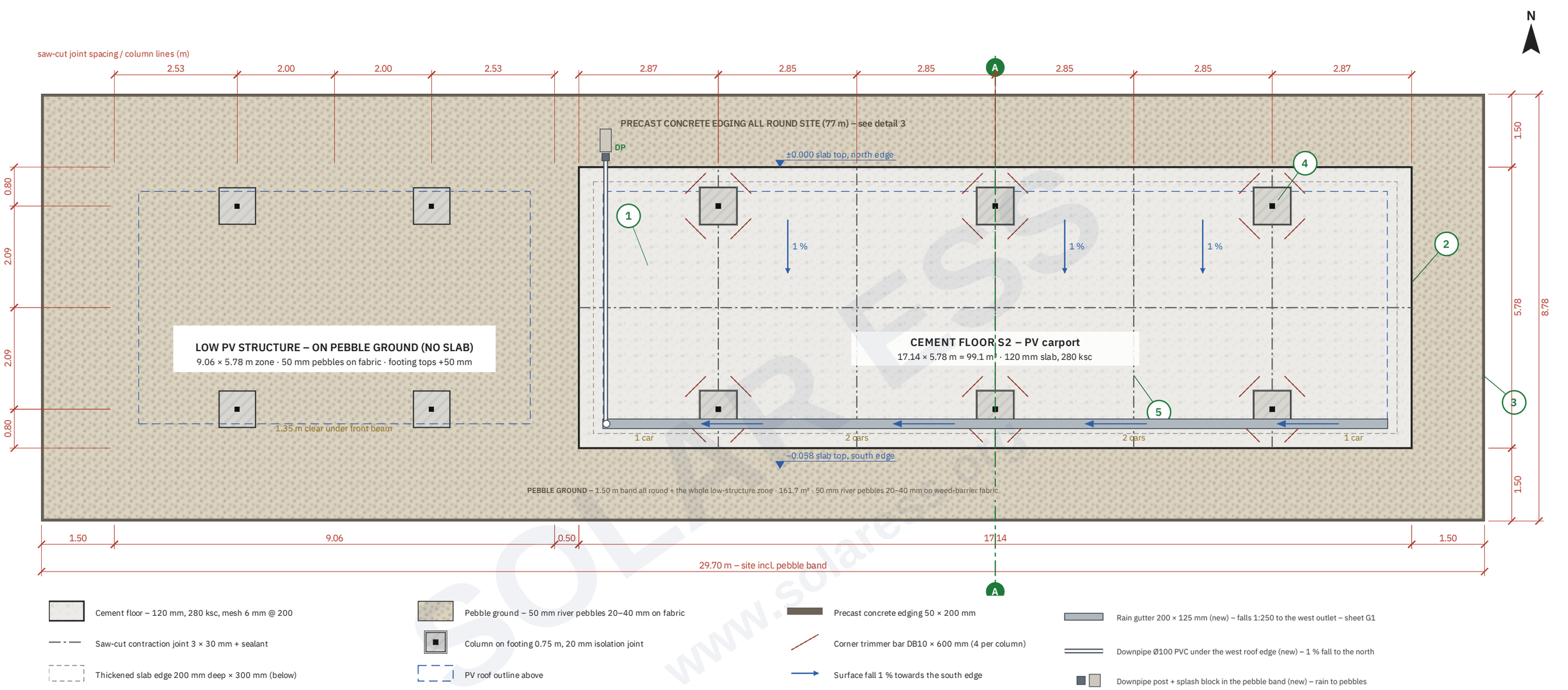
SOLAR ESS www.solaress.org

# Solar ESS Company Limited

**Floor plan – cement floor under the carport only, pebbles elsewhere** · setting-out, joints, falls and levels · scale 1:80 · Rev4

**F1**

Dimensions in metres (details in mm) · print A3 for scale
Preliminary – confirm with a licensed civil / structural engineer




## Floor schedule

| Zone | Size (m) | Area |
|---|---|---|
| S1 low PV structure – pebbles only | 9.06 × 5.78 | – (no slab) |
| S2 cement floor – PV carport | 17.14 × 5.78 | 99.1 m² |
| **Cement floor total** | | **99.1 m²** |
| Pebble ground (band + S1 zone + strip) | 29.70 × 8.78 – S2 slab | 161.7 m² |
| **Whole site** | **29.70 × 8.78** | **260.8 m²** |
| Saw-cut joints / isolation joints | | 41.5 / 19.0 m |
| Slab edge (thickened) / site edging | | 45.8 / 77.0 m |

## Setting out and levels

- The carport cement floor (S2) extends **0.50 m beyond its PV roof** on all sides; the low PV structure (S1) stands on **pebble ground – no slab**.
- Pebble band **1.50 m wide** all round, held by precast concrete edging on the site boundary; pebbles also cover the whole S1 zone.
- Datum **±0.000** = slab top at the north edge; set it **≥ 150 mm above existing ground** / access road (confirm on site).
- Slab falls **1 % to the south** (−0.058 at the south edge); rain from the PV roofs and slab soaks into the pebbles.
- Pebble surface **20 mm below the slab edge**; edging top 30 mm above the pebbles; S1 footing tops **50 mm above the pebbles**.

## Joints

- **Saw-cut joints** (carport slab only) on every column line and midway between them, plus one longitudinal joint at mid-width – bays ≤ 2.9 × 2.9 m (≤ 24 × slab thickness).
- Cut 3 mm × 30 mm deep **6–18 h after the pour**; later widen the top to 6 × 10 mm and seal with PU sealant (detail 5).
- Carport footing tops are cast flush with the slab and separated by a **20 mm isolation joint** (detail 4). Low-structure footings stand free in the pebbles – no joints.
- **DB10 × 600 mm corner bars** at the 4 corners of each carport footing stop cracks at re-entrant corners.

## Construction sequence

- 1 Clear, level and compact the whole site; set out columns and footings.
- 2 Cast the 10 footings – 6 carport footings up to slab level, 4 low-structure footings to +50 mm above the pebbles; anchor bolts with a template.
- 3 Carport slab only: crushed stone 150 mm in 2 layers (≥ 95 %) → sand 30 mm → PE sheet → mesh on chairs → edge forms.
- 4 Pour 280 ksc, broom finish, saw-cut, **wet-cure 7 days**; cars after 7 days, full load after 28 days.
- 5 Erect steel and PV roofs; seal joints.
- 6 Edging → fabric over the whole pebble area (incl. S1) → pebbles 50 mm; rake level.

# Solar ESS Company Limited

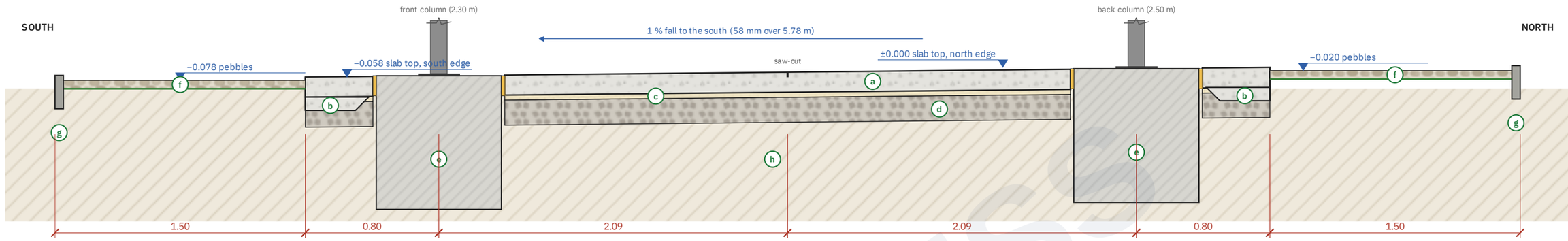
**Floor section and construction details** · section A–A through the PV carport middle frame · details 1–5

F2

Dimensions in metres (details in mm) · print A3 for scale
Preliminary – confirm with a licensed civil / structural engineer

## A–A Section A–A – north–south through the carport middle frame

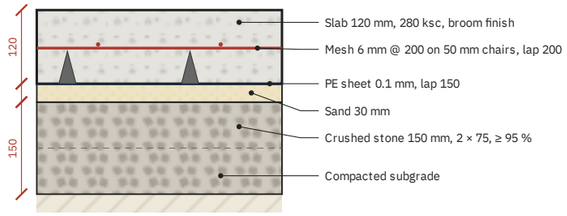
scale 1:25 · levels in metres · steel above floor level not shown




- (a) Cement floor 120 mm, ready-mix 280 ksc, welded mesh 6 mm @ 200 at mid-depth
- (b) Thickened slab edge 200 mm deep × 300 mm, mesh turned down (detail 2)
- (c) Sand 30 mm + PE sheet
- (d) Crushed stone 150 mm, 2 layers, ≥ 95 % compaction
- (e) Footing 0.75 × 0.75 × 0.80 m, cast up to slab level; 20 mm isolation joint (detail 4)
- (f) Pebble ground: 50 mm river pebbles 20–40 mm on weed-barrier fabric, 20 mm below slab
- (g) Precast concrete edging 50 × 200 mm on the site boundary (detail 3)
- (h) Levelled and compacted subgrade

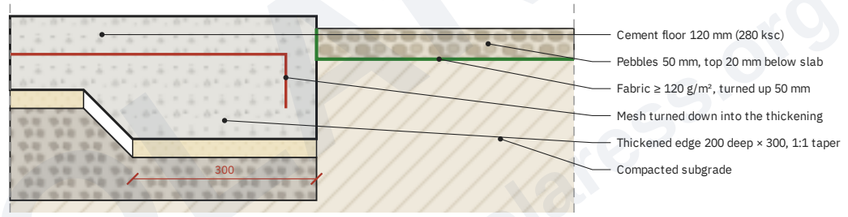
## 1 Typical cement floor build-up

scale 1:10



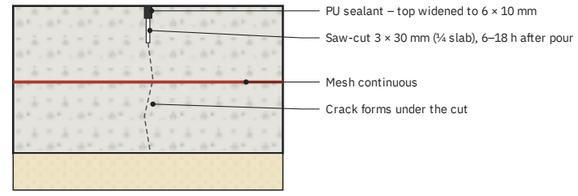

## 2 Slab edge to pebble ground

scale 1:10



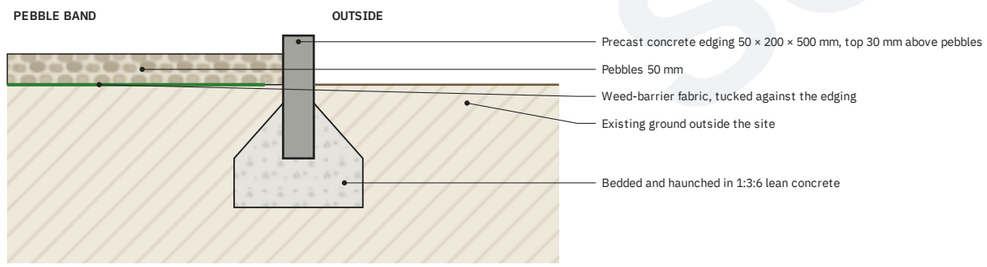

## 5 Saw-cut contraction joint

scale 1:5



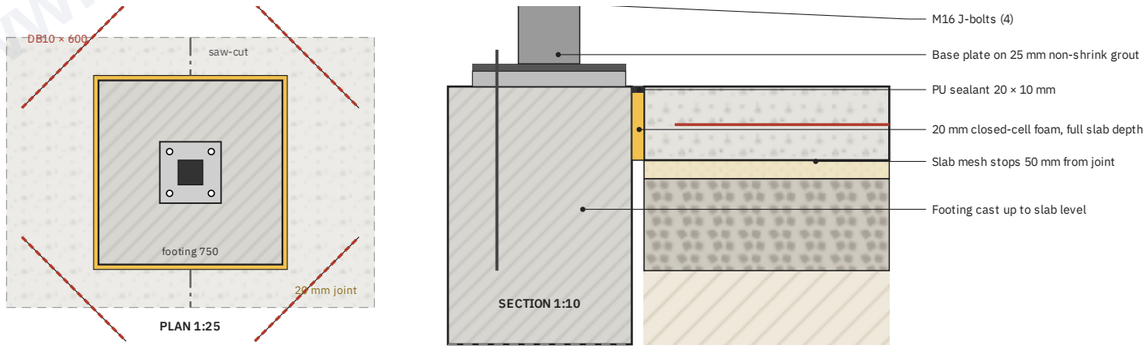

## 3 Site edge – precast concrete edging

scale 1:10




## 4 Column footing – isolation joint and corner bars

plan 1:25 · section 1:10




**Loads and use:** sedans and pickup trucks up to ≈ 3 t – not for 6-wheel trucks. Keep vehicles off for 7 days; full load after 28 days. **Curing:** keep the slab wet (hessian / water) for 7 days. **Tolerances:** slab surface ±5 mm under a 3 m straight-edge; no ponding. **Pebbles:** washed, rounded river pebbles 20–40 mm; top up yearly where thin. **Low PV structure (Rev4):** no slab under it – its 4 footings are cast to 50 mm above the pebbles, with the fabric turned up 50 mm around each footing.

# Solar ESS Company Limited

**Floor works – quantities (Rev4: cement floor under the carport only)** · cement floor 99.1 m² + pebble ground 161.7 m²

F3

Dimensions in metres (details in mm) · print A3 for scale
Preliminary – confirm with a licensed civil / structural engineer

## Floor works – reinforced cement floor under the carport (sedans & pickups) + pebble ground

| # | Item | Specification | Qty | Unit |
|---|---|---|---|---|
| 1 | Site preparation | Clear, level and compact the whole site (carport slab + pebble area) | 260.8 | m² |
| 2 | Crushed stone base | Crushed stone 150 mm, compacted in 2 layers (≥ 95 %), under the carport slab only | 17.8 | m³ |
| 3 | Sand bedding | Sand 30 mm levelling layer | 3 | m³ |
| 4 | Plastic sheet | Polyethylene sheet 0.1 mm under the slab, lap 150 mm | 109 | m² |
| 5 | Steel mesh | Welded steel mesh 6 mm @ 200 × 200 mm, on 50 mm chairs, lap 200 mm | 109 | m² |
| 6 | Concrete | Ready-mix 280 ksc (cube), 120 mm slab + thickened edges 200 mm, +5 % waste | 13.5 | m³ |
| 7 | Edge forms | Timber edge forms incl. thickened slab edge | 46 | m |
| 8 | Saw-cut joints | Saw-cut 3 × 30 mm within 6–18 h, widened 6 × 10 mm + PU sealant (layout F1) | 42 | m |
| 9 | Isolation joints | 20 mm closed-cell foam, full depth, + PU sealant around the 6 carport footings | 19 | m |
| 10 | Corner trimmer bars | DB10 SD40 × 600 mm, 4 per carport column (24 pcs = 14.4 m) | 2 | bar |
| 11 | Labour – slab | Compact base, place mesh, pour, broom finish, 7-day wet curing | 99 | m² |
| 12 | Weed-barrier fabric | Non-woven geotextile ≥ 120 g/m², lap 300 mm, turned up 50 mm at slab edge and footings | 178 | m² |
| 13 | Pebbles | River pebbles 20–40 mm, 50 mm layer (incl. +10 %) | 9 | m³ |
| 14 | Edging | Precast concrete edging 50 × 200 × 500 mm, bedded in lean concrete, around the site | 77 | m |
| 15 | Labour – pebbles | Lay fabric, spread and level pebbles (incl. around the low-structure footings) | 162 | m² |
| 16 | Transport | Ready-mix truck access, stone and pebble delivery | 1 | lot |

Quantities for the Rev4 design – cement floor under the carport only. Confirm quantities on site before ordering.

## What changed and why

- **Rev4:** cement floor under the PV carport only (S2, 99.1 m²); the low PV structure (S1) stands on pebbles – its 52.4 m² slab is deleted.
- **Slab items scaled to S2:** stone, sand, PE, mesh, concrete 20.5 → 13.5 m³, edge forms 76 → 46 m, saw-cuts 65 → 42 m, slab labour 152 → 99 m².
- **Isolation joints + corner bars** only at the 6 carport footings (32 → 19 m, 40 → 24 pcs).
- **Pebble ground** 109.3 → 161.7 m² (fabric 178 m², pebbles 9 m³); site prep, edging, transport unchanged.

## Areas and volumes

| | |
|---|---|
| Cement floor S2 (carport only) | 17.14 × 5.78 = 99.1 m² |
| Pebble ground | 161.7 m² |
| Concrete, net / to order (+5 %) | 12.7 / 13.5 m³ |
| Crushed stone to order (incl. compaction) | 17.8 m³ |
| Pebbles to order (+10 %) | 9.0 m³ |
| Saw-cut / isolation joints | 41.5 / 19.0 m |
| Slab edge / site edging | 45.8 / 77.0 m |

SOLAR ESS
www.solaress.org

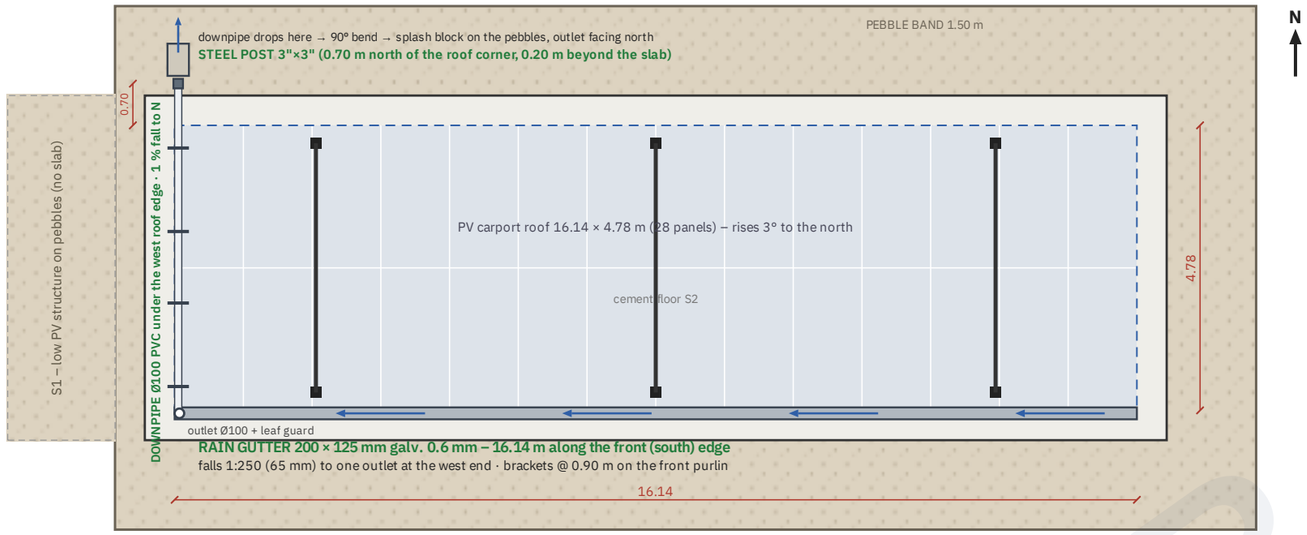
# Solar ESS Company Limited

**Rain gutter and downpipe – PV carport front edge** · plan, west elevation, details and bill of quantities

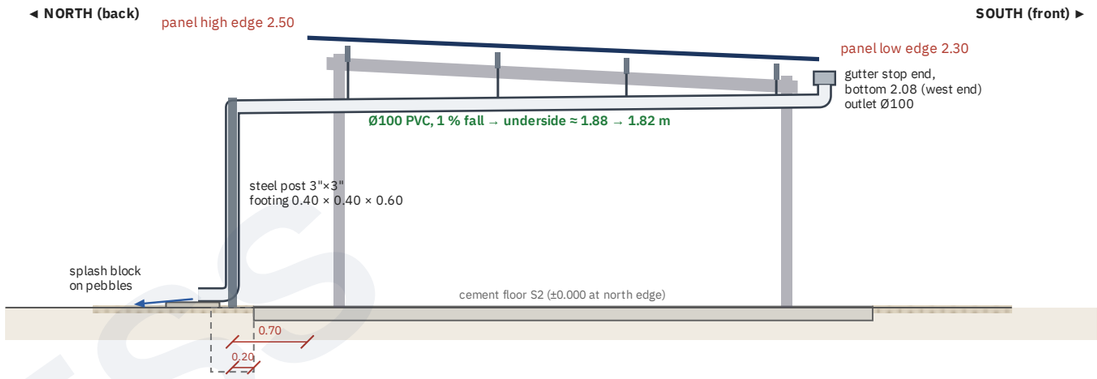
G1

Dimensions in metres (details in mm) · print A3 for scale
Preliminary – confirm with a licensed civil / structural engineer

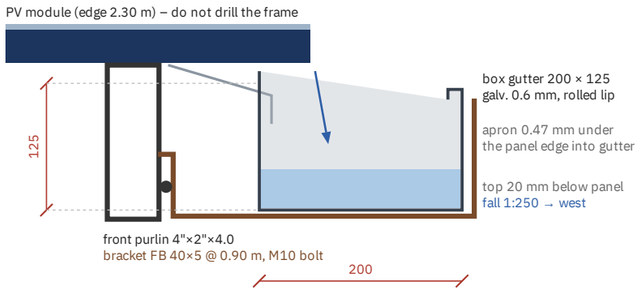
## PLAN – PV carport roof with gutter and downpipe route · scale ≈ 1:100 on A3



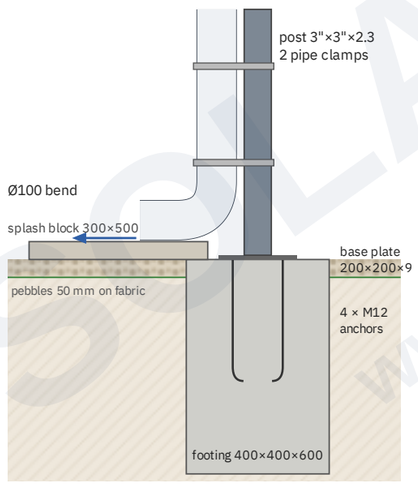

## WEST ELEVATION – downpipe route along the west roof edge · scale ≈ 1:50




## DETAIL 1 – gutter at the front edge · section, scale 1:5




## DETAIL 2 – post foot and splash block · 1:20




## Design basis

| | |
|---|---|
| Catchment (plan area of the carport roof) | **16.14 × 4.78 = 77.2 m²** |
| Design rainfall (short, intense storm – preliminary) | **150 mm/h** |
| Design flow Q = 77.2 × 150 / 3 600 | **3.2 L/s** |
| Gutter 200 × 125, fall 1:250, flowing 100 mm deep | **> 6 L/s – OK** |
| Downpipe Ø100 PVC at 1 % (horizontal run, full bore) | **≈ 7 L/s – OK** |
| Clear height under the pipe (west roof edge) | **≈ 1.88 → 1.82 m** |

## Bill of quantities – rain gutter + downpipe · quantities

| # | Item | Specification | Qty | Unit |
|---|---|---|---|---|
| 1 | Box gutter | Galvanised steel 0.6 mm, 200 × 125 mm, folded to order, 16.14 m + laps | 16.6 | m |
| 2 | Stop ends + outlet | 2 stop ends, 100 mm sump outlet with leaf guard at the west end | 1 | lot |
| 3 | Apron flashing | Zincalume 0.47 mm drip apron from panel edge into the gutter | 16.6 | m |
| 4 | Gutter brackets | Flat bar 40 × 5 mm galv. @ 0.90 m, bolted to the front purlin (M10) | 19 | pc |
| 5 | Downpipe | PVC Ø100 mm (4") class 8.5, 4 m lengths | 3 | length |
| 6 | PVC fittings | Outlet connector, 3 × 90° long bends, 2 × 45°, cleaning-eye tee, solvent cement | 1 | lot |
| 7 | Pipe hangers | Drop hangers from the 4 purlin tips + post clamps, stainless band clamps | 6 | set |
| 8 | Downpipe post | Tube 3"×3" × 2.3 mm, 6 m (2.4 m used, offcut for hangers) | 1 | length |
| 9 | Post plates | Base plate 200 × 200 × 9 mm, 4 × M12 anchors, pipe bracket at top | 1 | lot |
| 10 | Post footing | 0.40 × 0.40 × 0.60 m, concrete + 4 × DB12, dug in the pebble band | 1 | lot |
| 11 | Splash block | Precast concrete 300 × 500 mm on the pebbles, outlet facing north | 1 | pc |
| 12 | Paint | Zinc-rich primer + PU top coat on post, hangers and brackets | 1 | lot |
| 13 | Earth bonding | Gutter bonded to the carport earth, 10 mm² cable + lugs | 1 | lot |
| 14 | Labour | Gutter fabrication and fixing, downpipe, post and footing | 1 | lot |

## Notes

- **Route:** gutter along the front (south, low) edge, falling to the west (left) end; the Ø100 downpipe runs north under the west roof edge on 4 hangers from the purlin tips, then drops on its own post at the back (north-west) and discharges onto a splash block in the north pebble band – no water on the slab, nothing in the parking bays.
- **PV roof sealed (Rev5–7):** TS-10 seals, TX-10 crosses and clamp boots close every module gap and the row joint (sheets W1–W4), so all rain now runs off the front edge into this gutter – sized for the full roof catchment (≈ 76 m²).
- Do not screw into the module frames; fix brackets and apron to the front purlin, clear of the mid and end clamps. Bond the gutter to the carport earth.
- Pipe underside ≈ 1.82–1.88 m along the west edge – fine for cars, mark it with reflective tape for pedestrians. Clean the gutter and leaf guard before and during the rainy season.

SOLAR ESS
www.solaress.org

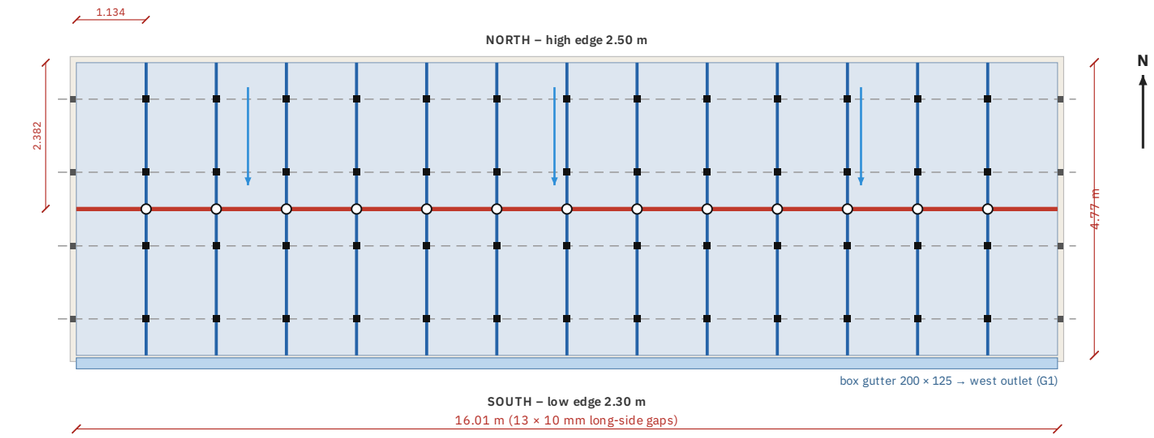
# Solar ESS Company Limited

**PV roof waterproofing – thin EPDM T-seals TS-10 in all 10 mm module gaps** · PV carport, 28 modules

**W1**

Dimensions in metres (details in mm) · print A3 for scale
Preliminary – confirm with a licensed civil / structural engineer

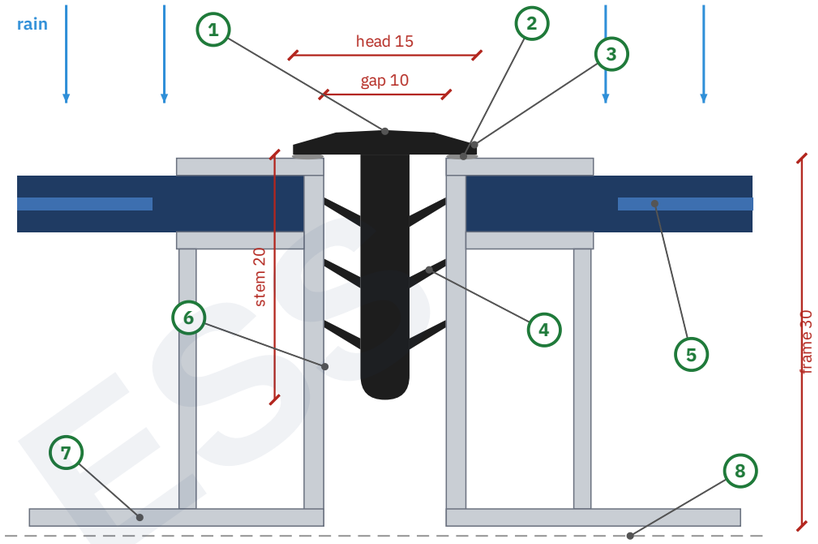
## PLAN – PV carport roof, joints to seal · scale 1:100 on A3




J1 – long-side joints N–S · 26 × 2.38 m · J2 – row joint E–W · 16.01 m in 14 segments · ○ 13 crossings – moulded cross TX-10 (W3)

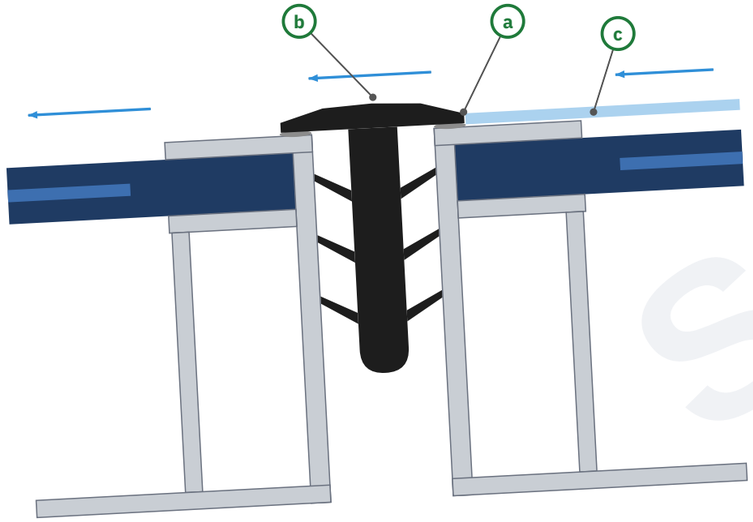
■ 52 mid clamps with boots TB-10 (D1, W4) · ■ 8 end clamps · purlins (indicative) · all rain runs south to the gutter

## SECTION 1 – J1 long-side joint with EPDM T-seal · scale 2:1 on A3 · typical between clamps



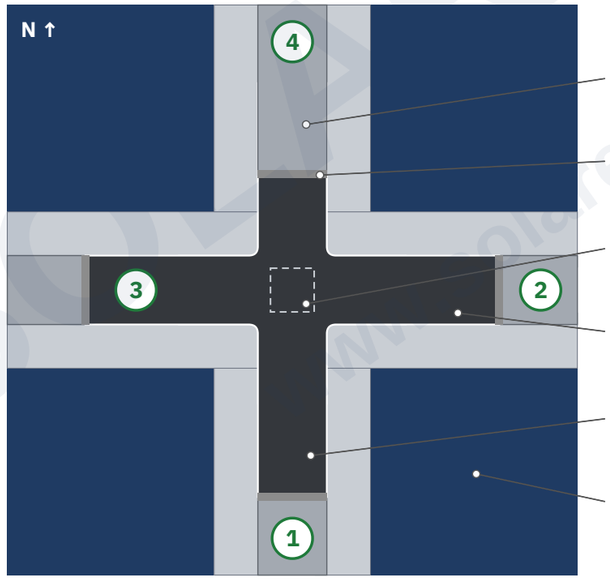

1. thin EPDM T-seal TS-10 (profile on W2): head 15 mm, stem 4 mm
2. MS-polymer bead under each wing – bonds and seals the head to the frame
3. wing 2.5 mm – rests on the aluminium frame top only
4. 3 pairs of thin fins press on both frame walls – grip and a second seal line
5. glass-glass laminate; the cells are well away from the seal – no shading
6. 10 mm gap (LONGi minimum) – between clamps the gap is clear full depth
7. frame bottom flange rests on the purlin at the clamp positions
8. purlin top (at clamp positions only – mid clamps and boots: D1, W4)

## SECTION 2 – J2 row joint, looking east · scale 2:1 · roof falls ≈ 3° to the south




- (a) feather edge 0.8 mm – water runs on and over the thin head
- (b) J2 seal, same TS-10 profile, in 14 segments between the moulded crosses
- (c) only ≈ 1 mm of water holds upslope of the head before it spills over – self-draining at ≈ 3°

## DETAIL 3 – crossing with moulded cross TX-10, plan · scale 3:4 on A3 · lay in order ① → ④ · details on W3




④ upper-row J1 lies on the N seat

MS-polymer at every lap end

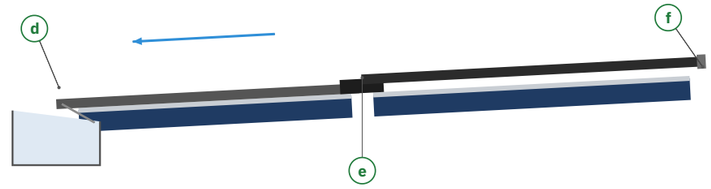
③ TX-10 cross – plug fills the corner

② J2 segments in the E / W sockets

① lower-row J1 in the S socket

glass – kept clear

## SECTION 3 – along one J1 joint · schematic, not to scale




- (d) south end: head runs 10 mm past the frame and drips onto the gutter apron (G1)
- (e) crossing with J2 – moulded cross TX-10 (W3); along every module length 2 mid clamps with boots TB-10 (W4)
- (f) north end: stem cavity closed with an MS-polymer plug

### How the seal keeps the roof dry

- Every 10 mm module gap gets a thin EPDM T-seal TS-10 (4 mm stem, 15 mm head) – 26 J1 joints along the slope and the J2 joint across it in 14 segments. At the 13 crossings a one-piece moulded cross TX-10 joins the two T-seals and plugs the corner void (W3).
- The 15 mm head rests only 2.5 mm on each frame edge – nothing touches the glass, no shading.
- Joints at the crosses are 20 mm moulded laps – a seat on the uphill arm, sockets under tongues on the others. The uphill piece is always on top and nothing is skived on site.
- **Rev7 – clamps:** aluminium mid clamps sit in the gap at the 52 purlin crossings, bolted into M8 rivet nuts in the purlin tops (D1). Each gets a moulded EPDM boot TB-10 that laps into the seal the same way as the crosses (W4).
- **Rev9:** 10 mm gaps on both the long sides and the short ends (LONGi minimum) – the Rev8 3 mm short-end joint is withdrawn. Roof 16.01 × 4.77 m. Frames, footings and floor unchanged; purlin cantilevers ≈ 2.30 m each end; gutter G1 can stay 16.14 m.
- Low PV structure (14 modules on pebbles): not sealed – optional (W2).

www.solaress.org

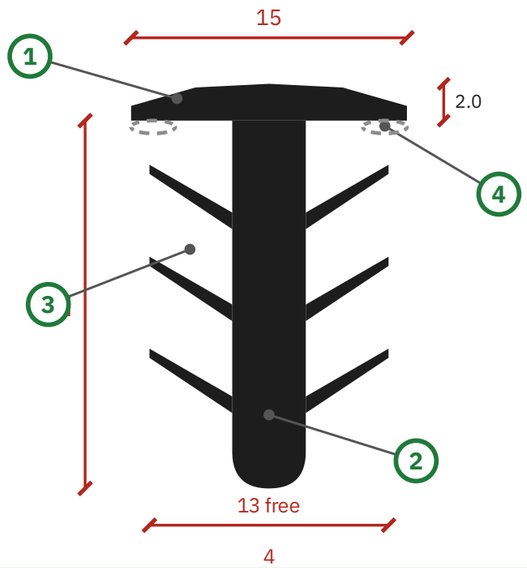
# Solar ESS Company Limited

**PV roof waterproofing – TS-10 profile, materials, quantities and installation** · PV carport, 28 modules

**W2**

Dimensions in metres (details in mm) · print A3 for scale
Preliminary – confirm with a licensed civil / structural engineer

## PROFILE TS-10 – thin EPDM T-seal to order · as extruded · scale 3:1 on A3 · dimensions in mm




1. head 15 mm – 2.0 mm thick at the centre, 0.8 mm feather edges
2. stem 4.0 mm thick, 20 mm deep, rounded tip
3. 3 pairs of fins at 5 / 10 / 15 mm – 0.8 → 0.5 mm thick, 30° upward, 13 mm tip to tip (fits gaps 9.5–12 mm)
4. MS-polymer bead ≈ 3 mm on each frame edge under the 2.5 mm wings (site-applied)

## Seal and materials specification

| | |
|---|---|
| **Profile** | TS-10 thin EPDM T-seal (this sheet) – head 15 mm, stem 4 mm × 20 mm, 3 pairs of fins, 13 mm free |
| **Fits** | module gap 10–11 mm (LONGi minimum 10 mm), frame depth ≥ 30 mm, flat frame edge ≥ 3 mm |
| **Material** | dense EPDM (not sponge), black, carbon-black UV-stabilised |
| **Hardness** | 60 ± 5 Shore A (soft enough for the thin fins) |
| **Weathering** | ozone- and UV-resistant – no cracks after ozone test (ISO 1431-1 / ASTM D1149) |
| **Temperature** | –40 … +120 °C (module frame up to ≈ 85 °C in sun) |
| **Strength** | tensile ≥ 8 MPa, elongation ≥ 300 %, compression set ≤ 35 % (22 h / 70 °C) |
| **Tolerances** | ISO 3302-1 class E2 · supplied in 50 m coils, straight cut |
| **Bedding** | MS-polymer bead ≈ 3 mm on each frame edge under the 2.5 mm wings – replaces butyl tape |
| **Cross junction** | TX-10 moulded cross, same EPDM compound – drawing and spec on W3 |
| **Clamp boot** | TB-10 moulded boot over each aluminium mid clamp – drawing and spec on W4 |
| **Sealant** | MS-polymer (hybrid), black, neutral cure – bonds EPDM, anodised Al and glass; no acetoxy silicone |
| **Cleaner** | isopropyl alcohol (IPA) ≥ 99 %, lint-free cloths |

## Quantities – PV carport, 28 modules

| # | Item | Basis | Qty | Unit |
|---|---|---|---|---|
| 1 | **Thin EPDM T-seal TS-10** | cutting list on W3: 74.0 m + 10 % | 85 | m |
| 1a | **Moulded cross junction TX-10** | 13 crossings + 2 spares (W3) | 15 | pc |
| 1b | **Moulded clamp boot TB-10** | 52 mid clamps + 4 spares (W4) | 56 | pc |
| 2 | **MS-polymer sealant, black, 290 ml** | beads under both wings (≈ 150 m) + 13 crosses + 52 boots + ends | 7 | cartridge |
| 3 | **IPA cleaner ≥ 99 %** | frame edges and gap walls | 1 | litre |
| 4 | **Lint-free cloths** | also to wipe sealant squeeze-out off the glass | 2 | pack |
| 5 | **Gap spacer blocks 10 mm** | plastic, to set the gaps before final clamping | 20 | pc |
| 6 | **Hand roller + plastic wedge** | silicone roller 20 mm; plastic wedge to push the stem in – no metal tools on glass | 1 | set |
| **Option – low PV structure, 14 modules (only if the area under it must stay dry)** | | | | |
| 01 | Thin EPDM T-seal TS-10 | A–F × 6 + J2 7 segments (W3 lengths) + 10 % | 39 | m |
| 02 | MS-polymer sealant, black, 290 ml | beads + 4 laps at each of 6 crosses + 14 ends | 3 | cartridge |
| 03 | Moulded cross junction TX-10 | 6 crossings + 1 spare | 7 | pc |
| 04 | Moulded clamp boot TB-10 | 24 mid clamps + 2 spares | 26 | pc |

## Installation sequence

1. **Set the gaps** to 10–11 mm with the spacer blocks before final clamping – never under 10 mm (LONGi). Keep module cables and connectors clipped under the frames, out of the gaps.
2. **Test-fit a 1 m sample** of TS-10 in 3 gaps: the stem must go in with a firm push, the fins must grip, and the head must lie flat on both frame edges.
3. **Clean** the frame edges and gap walls with IPA; let them dry. Work in dry weather, frame temperature 10–45 °C.
4. **Cut lengths relaxed.** Unroll the coil and let it rest flat 24 h in the shade. Cut pieces A–F and J2 to the W3 cutting list – straight cuts only; trim the stem off the last 20 mm where the list says so.
5. **Work uphill along every joint**: seal piece → boot (W4) or cross (W3) → next seal piece, each lapping onto the one below. MS-polymer bead on both frame tops, press the stem in with the wedge, roll the head. Start at the south edge (pieces F run 10 mm past the frame).
6. **Clamps first:** torque all mid and end clamps (12–16 N·m, D1) before any boot goes on. Lay the J2 segments before each cross. Plug the north and roof-end ends with MS-polymer.
7. **Wipe** any sealant squeeze-out off the glass before it skins (≈ 10 min).
8. **Water test** after 24 h: hose each joint from the north edge for 10 min and check the underside is dry; then run the full roof and check that the gutter takes all the water.

## Checks and notes

- **Module gap:** the LONGi Hi-MO X10 installation manual (V19) requires at least 10 mm between adjoining modules. This design keeps 10–11 mm – do not close the gaps further.
- **No shading:** the 2.5 mm wings stay on the frame edges, well clear of the cells. The bifacial rear side stays open to the floor.
- **Roof size (Rev6):** 14 × 1 134 + 13 × 10 = 16.01 m long, 2 × 2 382 + 10 = 4.77 m wide (was 16.14 × 4.78 m with 20 mm gaps). Frames, footings and floor unchanged; purlin cantilevers ≈ 2.30 m each end.
- **Gutter G1:** all rain now leaves at the front edge; G1 is sized for the full roof (≈ 76 m²) – OK, and can stay 16.14 m long.
- **Pitch ≈ 3°:** the thin J2 head holds back only ≈ 1 mm of water before it spills over; clean it with every panel wash (every 6 months).
- **Frame drainage holes** in the module frame corners must stay open – do not fill them with sealant.
- **No load on glass:** the seal is not structural. Do not walk or kneel on modules – work from a mobile scaffold or ladders; modules are live in daylight.
- **Warranty:** send D1 and W1–W4 to the LONGi distributor to confirm that the frame-edge seals are acceptable under the module warranty.
- **Supply:** TS-10 is a small custom extrusion – ask extruders for their nearest standard T-seal die for 10 mm gaps first; TX-10 and TB-10 are moulded. Order samples of all three, and fit the TB-10 sample over the actual mid clamp.

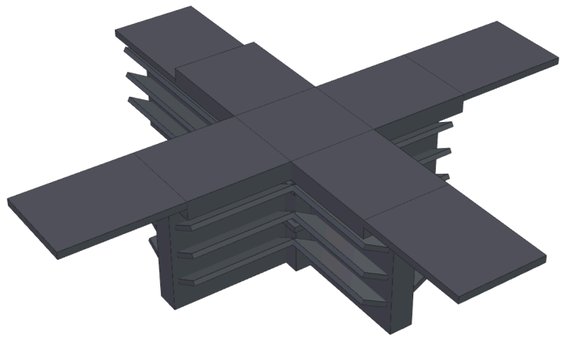
# Solar ESS Company Limited

**PV roof waterproofing – TX-10 moulded cross where two thin T-seals meet** · 10 mm gaps both ways · 13 crossings

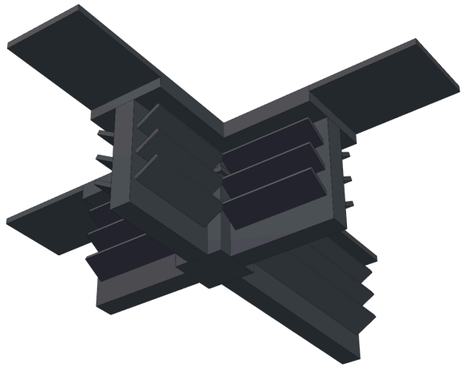
W3

Dimensions in metres (details in mm) · print A3 for scale
Preliminary – confirm with a licensed civil / structural engineer

## TX-10 – moulded EPDM cross junction · 3D views, not to scale

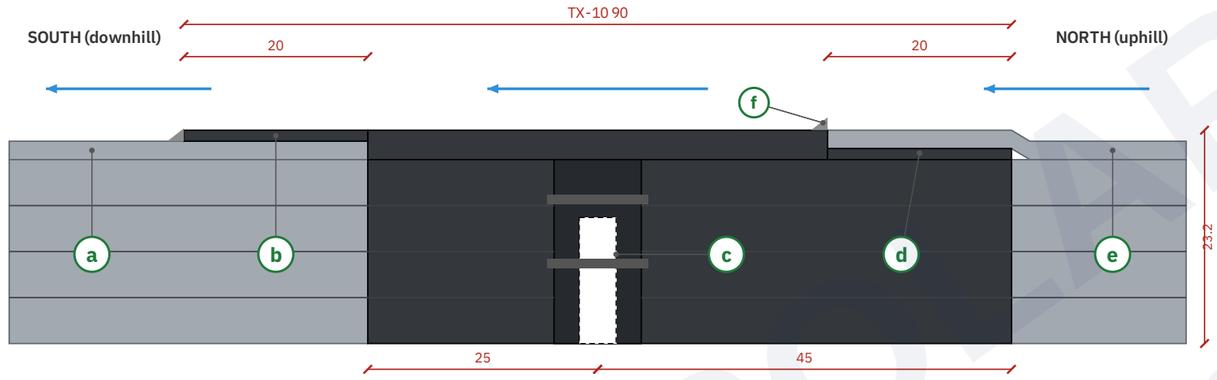


from above – the N arm (back left) has the 2 mm seat; the other three arms end in tongues over sockets



from below – centre plug with 2 ribs, 4 mm stems and thin fins under all four arms

## SECTION A–A – along the J1 line through the cross · scale 1.5:1 on A3 · roof falls to the south (left) · dashed line = frame top level




(a) lower-row J1 (TS-10): straight cut; head slides 20 mm into the S socket, stem butts the cross stem at 25 mm

(b) S tongue 1.2 mm lies over it – uphill piece on top, MS-polymer in the socket

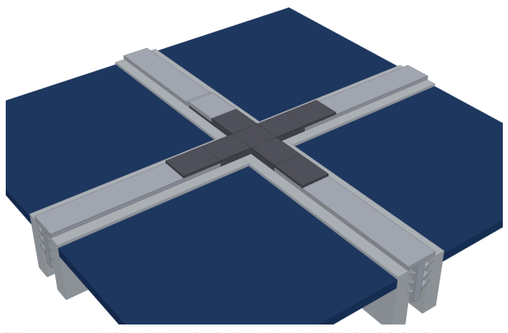
(c) centre plug with 2 ribs and a hollow core, pressed into the corner void

(d) N seat moulded 2.0 mm down – the upper-row J1 head lies on it, flush with the cross top

(e) upper-row J1 (TS-10): stem trimmed off its last 20 mm, butts the N arm stem at 45 mm

(f) MS-polymer fillet at each lap end line – no step faces uphill anywhere

## Fitted at a crossing · 3D view



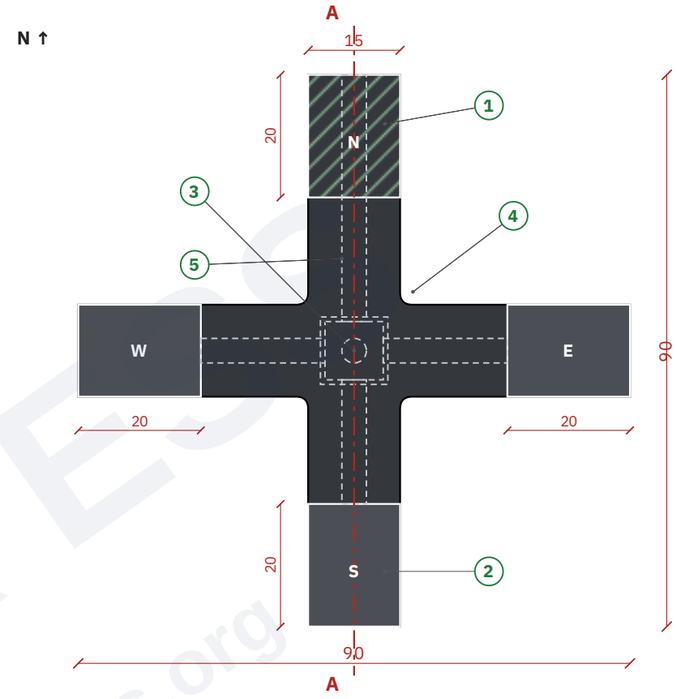
light grey = straight TS-10 seals, dark = TX-10. North is back left: the upper-row J1 lies on the cross seat; the other three seals slide into the cross sockets.

## Cutting list – TS-10 pieces for the carport (Rev9)

| # | Piece | Qty | Stem | Head | Ends |
|---|---|---|---|---|---|
| **A** | Upper row – north piece | 13 | 547 | 567 | N: flush + plug · S: on boot seat, stem trimmed 20 |
| **B** | Upper row – middle piece | 13 | 1 116 | 1 136 | N: into boot socket · S: on boot seat, stem trimmed 20 |
| **C** | Upper row – south piece | 13 | 527 | 547 | N: into boot socket · S: on TX-10 N seat, stem trimmed 20 |
| **D** | Lower row – north piece | 13 | 527 | 547 | N: into TX-10 S socket · S: on boot seat, stem trimmed 20 |
| **E** | Lower row – middle piece | 13 | 1 116 | 1 136 | N: into boot socket · S: on boot seat, stem trimmed 20 |
| **F** | Lower row – south piece | 13 | 567 | 577 | N: into boot socket · S: 10 mm past the frame (drip) |
| **J2** | J2 middle segments | 12 | 1 094 | 1 094 | straight cuts into TX-10 E / W sockets |
| **J2e** | J2 end segments | 2 | 1 114 | 1 114 | cross end straight · roof end flush + plug |
| **TX** | TX-10 moulded cross | 13 + 2 | – | – | W3 |
| **TB** | TB-10 clamp boot | 52 + 4 | – | – | W4 |

Lengths in mm, per J1 joint: A–C upper row, D–F lower row, with mid clamps (boots) at 595 mm from each module end. Total TS-10 = 74.0 m (+10 % → 85 m, W2).

## PLAN – TX-10 · scale 1:1 on A3 · dimensions in mm




(1) N arm (uphill): 20 mm seat moulded 2.0 mm down into the top – the upper-row J1 head lies on it, flush

(2) S, E and W arms: 20 mm sockets under 1.2 mm tongues – the incoming seal heads slide in; straight cuts only, no skiving

(3) centre plug 9.5 × 9.5 × 20 mm with 2 ribs (11 × 11) fills the 10 × 10 mm void where the 4 module corners meet

(4) inside corners R2 – the head stays on the frame edges

(5) stems 4 mm with 3 pairs of fins under every arm (dashed): N arm stem to 45 mm, S / E / W arm stems to 25 mm

## TX-10 specification

| | |
|---|---|
| **Part** | TX-10 cross junction – one piece, compression-moulded EPDM |
| **Compound** | same as TS-10: dense EPDM 60 ± 5 Shore A, black, UV + ozone resistant, –40 … +120 °C |
| **Size** | 90 × 90 mm, arms 15 mm wide, head 3.2 mm (1.2 mm above the straight seals), total depth 23.2 mm |
| **Centre plug** | 9.5 × 9.5 × 20 mm, 2 ribs 11 × 11 × 1.0 mm, hollow core Ø 4 × 12 mm, 1.5 mm lead-in chamfer |
| **Stems / fins** | 4 mm core, 3 pairs of fins at 5 / 10 / 15 mm as TS-10 – N arm to 45 mm, S / E / W arms to 25 mm |
| **Lap zones** | 20 mm at every arm end – N: seat 2.0 mm deep with an end ramp; S / E / W: sockets 2.0 × 15.5 mm under 1.2 mm tongues |
| **Weight** | ≈ 20 g |
| **Tolerances** | ISO 3302-1 class M2 · steel 2-part mould with side cores – fins flex out on demoulding |
| **Approval** | 5 first-off samples test-fitted on a real module crossing before the series is moulded |

## Fitting a crossing

1. Check the 4 module corners: gaps 10–11 mm (never under 10), and the upper- and lower-row J1 gaps in line within ± 1.5 mm (string line).
2. Clean the frame edges around the crossing with IPA.
3. Lay the lower-row J1 and both J2 segments first – straight-cut ends stopping 25 mm from the crossing centre.
4. MS-polymer beads on the frame edges under the cross wings and a thin bead inside the 3 sockets.
5. Press the centre plug into the corner void first, then the 4 arm stems; the 3 sockets slide over the incoming seal heads.
6. MS-polymer on the N seat; lay the upper-row J1 (stem trimmed off its last 20 mm) onto it.
7. Tool small MS-polymer fillets along the 4 lap end lines; wipe the glass clean.
8. Hose-test the crossing for 10 min from uphill – the underside must stay dry.

SOLAR ESS www.solaress.org

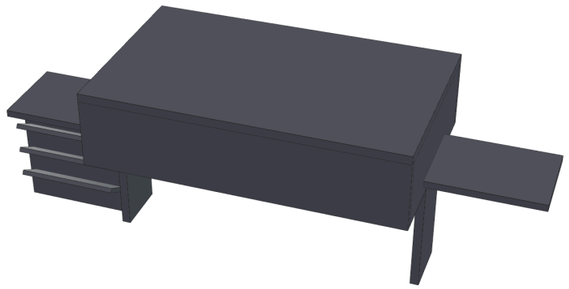
# Solar ESS Company Limited

**PV roof waterproofing – TB-10 moulded boot over each aluminium mid clamp** · 52 on the PV carport

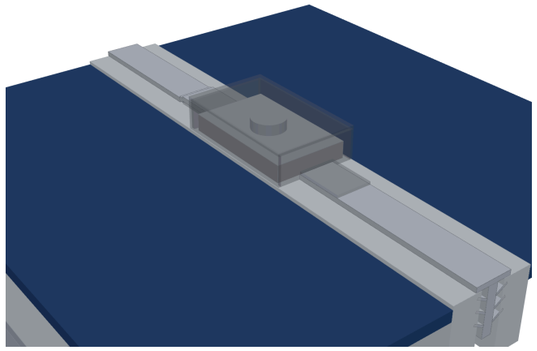
W4

Dimensions in metres (details in mm) · print A3 for scale
Preliminary – confirm with a licensed civil / structural engineer

## TB-10 – moulded EPDM clamp boot · 3D views, not to scale

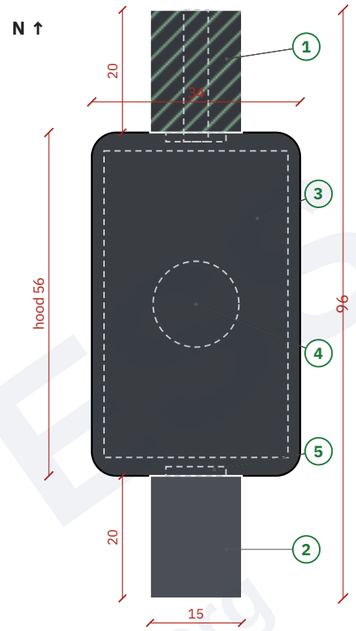


the boot: hood over the clamp, seat on the uphill arm (left, with finned stem), tongue over the socket on the downhill arm (right)



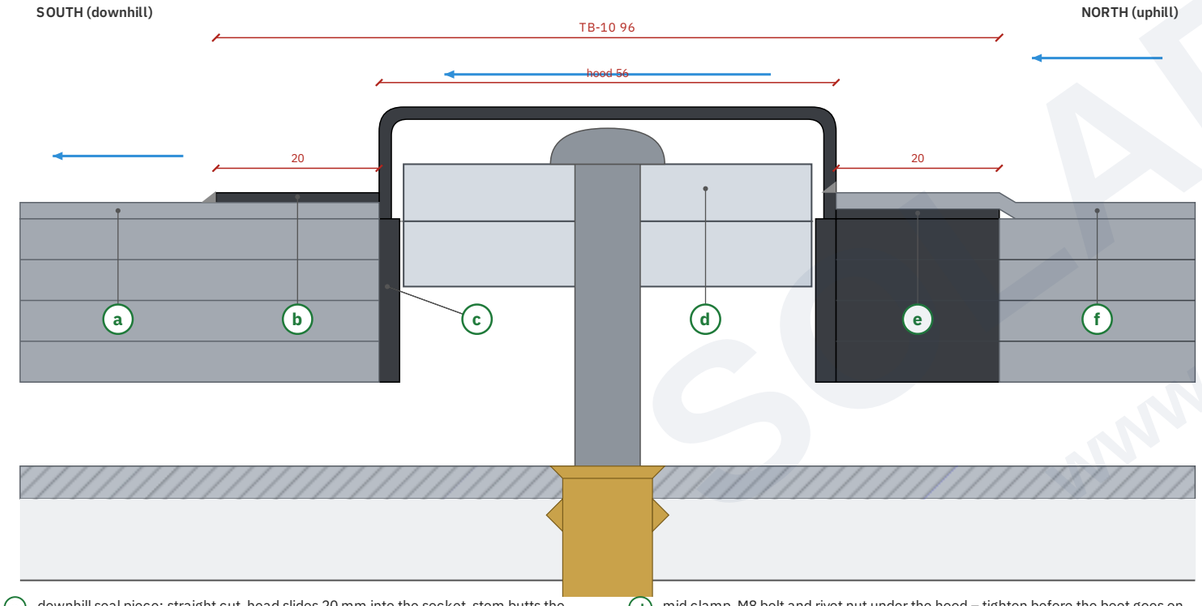
fitted: boot shown see-through over the mid clamp and M8 bolt; the uphill seal (back) lies on the seat, the downhill seal (front) runs into the socket

## PLAN – TB-10 · scale 1:1 on A3




1. uphill (north) arm: 20 mm seat moulded 2.0 mm down – the uphill seal lies on it, flush
2. downhill (south) arm: 20 mm socket under a 1.2 mm tongue – the downhill seal slides in
3. hood 56 × 34 × 14 mm, R4 corners – covers the 50 × 30 clamp and the bolt head; skirt stays on the 12 mm frame tops
4. clamp and M8 bolt inside (dashed)
5. end dams drop into the 10 mm gap at both ends of the hood; the uphill arm carries a 4 mm finned stem

## SECTION – along the joint through a mid clamp and its boot · scale 1.33:1 on A3 · roof falls to the south (left) · dashed line = frame top




a. downhill seal piece: straight cut, head slides 20 mm into the socket, stem butts the end dam
b. 1.2 mm tongue of the boot over it – uphill piece on top, MS-polymer in the socket
c. end dams in the gap at both ends of the hood
d. mid clamp, M8 bolt and rivet nut under the hood – tighten before the boot goes on
e. seat 2.0 mm down; finned stem under it
f. uphill seal piece: stem trimmed off its last 20 mm, head lies on the seat flush with the boot arm

## TB-10 specification

| | |
|---|---|
| **Part** | TB-10 clamp boot – one piece, compression-moulded EPDM, same compound as TS-10 (60 ± 5 Shore A, UV + ozone resistant) |
| **Size** | 96 × 34 mm overall; hood 56 × 34 × 14 mm high, walls 1.0 mm sides / 1.5 mm roof and ends (inside 53 × 32 × 12.5); arms 15 mm wide |
| **Fits** | generic aluminium mid clamp 50 × 30 × 7 mm with an M8 button head ≤ 11.5 mm above the frame, frame tops ≥ 12 mm – confirm on the chosen clamp and module, adjust the hood |
| **Laps** | uphill arm: 20 mm seat 2.0 mm deep + finned 4 mm stem · downhill arm: 20 mm socket 2.0 × 15.5 mm under a 1.2 mm tongue |
| **Gap dams** | end dams 9.8 × 1.5 × 20 mm at both ends of the hood, in the 10 mm gap |
| **Bedding** | MS-polymer bead under the hood skirt on both frames and in the socket |
| **Weight** | ≈ 25 g |
| **Access** | to re-torque a clamp: cut the bead, lift the boot, re-bed with fresh MS-polymer |
| **Approval** | 5 first-off samples fitted over the real clamp before the series is moulded |

## Fitting a boot

1. Torque the mid clamp to 12–16 N·m and check the earthing first.
2. Lay the downhill seal piece; its straight-cut end stops 28 mm from the clamp centre.
3. MS-polymer bead on the frame tops along the hood skirt lines and a thin bead in the socket.
4. Press the boot over the clamp: the tongue slides over the downhill seal head, the end dams drop into the gap.
5. MS-polymer on the seat; lay the uphill seal (stem trimmed off its last 20 mm) onto it.
6. Tool fillets at both lap end lines; wipe the glass clean before the sealant skins.

SOLAR ESS www.solaress.org